
ADVANTEST®

ADVANTEST CORPORATION

ASX/U-51

SYSTEST TESTUTL ATLworks

UPDATE NOTICE

REV 6.00C

MANUAL NUMBER 8262527-55

Applicable Products

PASX51-92****
PASX51-91****
PASX51-85****
PASX51-81****
PASX51-75****
PASX51-71****
PASX51-70****
PASX51-36****

PREFACE

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RECORD OF REVISIONS

Manual Rev	Date	Remarks	Manual Rev	Date	Remarks
43a	Jan 10/01	(REV 5.04)	51	Dec 26/01	(REV 6.00-2)
44	Feb 20/01	(REV 5.04A)	52	Jan 9/02	(REV 6.00A)
45	Apr 3/01	Obsolete	53	Feb 25/02	(REV 6.00A1)
45a	Apr 5/01	Obsolete	54	Jul 15/02	(REV 6.00B)
45b	Apr 6/01	(REV 5.04A1)	55	Nov 1/02	(REV 6.00C)
46	Apr 7/01	(REV 5.04A2)			
47	Apr 9/01	Obsolete			
47a	Sep 25/01	(REV 5.04A3)			
48	Apr 10/01	(REV 5.05)			
49	Aug 20/01	Obsolete			
49a	Sep 25/01	(REV 6.00)			
50	Oct 18/01	(REV 6.00-1)			

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1. SUMMARY OF REVISION

- (1) Revision number
 - ASX/U-51 REV 6.00C
- (2) Applicable Systems
 - T5592
 - T5591/T5591R
 - T5586/T5585
 - T5581/T5581H
 - T5581P/T5571P
 - T5376/T5375
 - T5371
 - T5336/T5311
- (3) Related manual revised

• ATL-51 TEST PLAN PROGRAM REFERENCE MANUAL	8431498-02
• ATL-51 SOCKET PROGRAM REFERENCE MANUAL	8256293-17
• T5500 SERIES TESTER UTILITY PROGRAM REFERENCE MANUAL	8256295-33
• T5500 SERIES CONTROLLER COMMAND REFERENCE MANUAL	8256297-21
• ATLworks TOOL USER'S MANUAL	8311174-11
• ASX/U-51 HYBRID CONTROLLER USER'S MANUAL	8409598-05
• ASX/U-51 HYBRID CONTROLLER REFERENCE MANUAL	8409600-04
• ASX/U-51 BASIC SYSTEM SOFTWARE SOFTWARE SPECIFICATIONS	8256273-11a
- (4) Reason of revision
 - Addition of function
 - Improvement of existing functions

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1. SUMMARY OF REVISION

(5) Products revised

Products revised in this revision are shown below:

Product code	Product name		
PASX51-92****	SYSTEST51SU	TESTUTL51SU	ATW51SU
PASX51-91****			
PASX51-85****			
PASX51-81****			
PASX51-75****			
PASX51-71****			
PASX51-70****			
PASX51-36****			

(6) List of revisions

The revision list shows the revision correspondence of the basic software the SYSTEST51SU, TESTUTL51SU and ATW51SU of ASX/U-51.

<Explanation of the revisions' list>

(a) "*" shows the product revised in the revision.

(b) "–" shows the product that does not exist to the whole revision.

ASX/U-51				
ASX/U-51 Overall Revision	Revision for each product			SunOS Revision
	SYSTEST51SU	TESTUTL51SU	ATW51SU	
5.04A	5.04A *	5.04A *	–	Solaris 2.5.1 Solaris 7
5.04A1	5.04A1 *	5.04A1 *	–	Solaris 2.5.1 Solaris 7
5.04A2	5.04A2 *	5.04A1	–	Solaris 2.5.1 Solaris 7
5.04A3	5.04A3 *	5.04A1	–	Solaris 2.5.1 Solaris 7
5.04A4	5.04A4 *	5.04A2 *	–	Solaris 2.5.1 Solaris 7
5.05(Note1)	5.05 *	5.05 *	5.05 *	Solaris 8

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1. SUMMARY OF REVISION

ASX/U-51				
ASX/U-51 Overall Revision	Revision for each product			SunOS Revision
	SYSTEST51SU	TESTUTL51SU	ATW51SU	
6.00	6.00 *	6.00 *	6.00 *	Solaris 2.5.1 Solaris 7 Solaris 8
6.00-1	6.00-1 *	6.00-1 *	6.00-1 *	Solaris 2.5.1 Solaris 7 Solaris 8
6.00-2	6.00-2 *	6.00-1	6.00-1	Solaris 2.5.1 Solaris 7 Solaris 8
6.00A	6.00A *	6.00A *	6.00A *	Solaris 2.5.1 Solaris 7 Solaris 8
6.00A1	6.00A1 *	6.00A1 *	6.00A	Solaris 2.5.1 Solaris 7 Solaris 8
6.00B	6.00B *	6.00B *	6.00B *	Solaris 2.5.1 Solaris 7 Solaris 8
6.00C	6.00C *	6.00C *	6.00C *	Solaris 2.5.1 Solaris 7 Solaris 8

Note1: REV 5.05 is used for the T5375.

2. ENHANCED FUNCTIONS

2.1 Fail Memory

All the DUT fails are captured into the fail memory at one time, even when the test rate is 7 ns or more. For more information, refer to item, SELECT AFM CONFIG statement in “ATL-51 TEST PLAN PROGRAM REFERENCE MANUAL.”

The following program has been updated according to this added function.

- Tester driver controller (stest.sys)

Relevant systems: T5376 and T5375

2.2 2-dimension Shmoo Plot (SHM2)

VSIM, HVSIM, and ISVM have been added to the target unit for FUNCTION SHMOO and PARALLEL SHMOO.

For more information, refer to “T5500 SERIES TESTER UTILITY PROGRAM REFERENCE MANUAL.”

Relevant systems: T5592, T5591, T5591R, T5586, T5585, T5581, T5581H, T5581P, T5571P, T5376, T5375, T5371, T5336, and T5311

2.3 AFM Two-speed Mode Address Setting Utility (FMADRSET)

This utility is used to assign the AFM two-speed mode address to the AFM instead of the AFM ADDRESS statement. The device burst address progress exists in several combination depending on the burst type, burst length, and start address. However, two-speed mode setting of the AFM ADDRESS statement cannot be used with all the combination. (*1)

This utility can be used with all the combination by using the prescrambler to generate the burst address. This utility can analysis fails which could not be specified by the AFM ADDRESS statement.

(*1) The AFM ADDRESS statement cannot specify fails under the following conditions:

- The burst length is 4 or more, the start address is an odd number, such as 1, 3, 5, or 7, and the burst type is sequential.

For more information, refer to “T5500 SERIES TESTER UTILITY PROGRAM REFERENCE MANUAL.”

Relevant systems: T5592, T5591, T5591R, T5586, T5585, T5376, T5375, and T5371

2.4 Controlling the Handler by using the GPIB Interface (/STHAND)

This program can be used with the M6542AD.

For more information, refer to “T5500 SERIES TESTER UTILITY PROGRAM REFERENCE MANUAL.”

Relevant systems: T5592, T5591, T5591R, T5586, T5585, T5581, T5581H, T5581P, T5571P, T5376, T5375, T5371, T5336, and T5311

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2.4 Controlling the Handler by using the GPIB Interface (/STHAND)

Relevant handler: M6542AD

Relevant handler system software: M6542AD SYSTEM SOFTWARE REV 1.00

3. IMPROVED ITEMS

3.1 Tester Driver Controller (stest.sys)

3.1.1 AFM ACTION Statement

When the AFM two-speed mode was specified by using the <Xpx-qx, Ypy-qy> setting and the AFM ACTION statement was specified in a mode which used strobes, no runtime error occurred.

Improved ➡ A runtime error (code=12324) now occurs.

Relevant revisions: REV 5.05 through REV 6.00B

Relevant systems: T5592, T5376, and T5375

3.1.2 READ AFM CONFIG Data Statement

When the SELECT AFM CONFIG DOUBLE was specified; if the value, which exceeds the number of groups that was read by using the READ AFM CONFIG COUNT statement; was specified in the argument, which was used to specify which group information number in the READ AFM CONFIG DATA statement was to be read; no runtime error occurred.

Improved ➡ A runtime error (code=3111) now occurs.

Relevant revisions: REV 5.05 through REV 6.00B

Relevant systems: T5376 and T5375

3.1.3 READ AFM Fail Count Statement

When conditions below, (a) through (c), were satisfied, if the DUT setting in the READ AFM fail count statement was set to ALL and the number of fails in all the DUTs were counted, the number of counted fails doubled.

Conditions

- (a) A socket program is used, which does not use pins 97 through 114.
- (b) DOUBLE is specified in the SELECT AFM CONFIG statement.
- (c) ALL or CONFIG is specified in the AFM MULTIMODE statement.

Improved ➡ This problem has been fixed.

Relevant revisions: REV 5.05 through REV 6.00B

Relevant systems: T5376 and T5375

3.1.4 Tester Driver Controller Start Process

When the tester driver controller started, the settings shown below were set to PPS 1 to 4 and 97 to 100 in the ROF state. In this situation, the Kelvin alarm sometimes occurred and this resulted in a runtime error (code=3521).

Applied voltage: 0 V
 Voltage apply range: R (5V)
 Measurement range: M (800MA)
 Current clamp value: Ip=800 mA, Im=-800 mA

The same error occurred when the following command was executed or buttons were pressed:

- CTRL + C command
- Control box RESET switch
- Operation panel Reset button
- Control Panel (ATLworks) Reset button
- ASX/Debugger (ATLworks) Reset button

Improved ➡ PPS 1 to 4 and 97 to 100 are in the OPEN state and the Kelvin alarm does not occur.

Relevant revisions: REV 2.00 through REV 6.00B

Relevant systems: T5592, T5591, T5591R, T5586, T5585, T5581P, T5571P, T5376, T5375, T5371, T5336, and T5311

3.1.5 END, STOP, and SET DECISION Statements

When a parallel test socket program, which used two stations, was used, the fail counter on station 2 was not counted when both of the following conditions were met:

- The program name on station 1 executed last was different from the program name to be executed next.
- All DUTs on station 1 were non-target DUTs.

Improved ➡ The fail counter can be counted.

Relevant revisions: REV 5.03 through REV 6.00B

Relevant systems: T5592, T5586, T5585, T5376, T5375, and T5371

3.1.6 RESET AFM Statement

When RESET AFM MEMORY was executed by using AFM PGSEL=MAIN, SUB and AFM RWSEL=MAIN, both MAIN and SUB mark memories are cleared.

Improved ➡ The SUB mark memory is not cleared.

Relevant revisions: REV 6.00 through REV 6.00B

Relevant systems: T5586 and T5585

3.2 Tester Utility Program

3.2.1 Pin Monitor (/PM)

When, in the pin statement, XOR was specified in the waveform mode, and BCLK and CCLK were specified in the clock, displaying the pin connections displayed 00213C in the waveform mode.

Improved ➡ XOR is now displayed.

Relevant revisions: REV 5.02 through REV 6.00B

Relevant system: T5592

3.2.2 Extension Pin Monitor (/EPM)

- (1) When, in the pin statement, XOR was set to the waveform mode, and BCLK and CCLK were specified in the clock, displaying the pin connections displayed 00213C in the waveform mode.

Improved ➡ XOR is now displayed.

Relevant revisions: REV 5.02 through REV 6.00B

Relevant system: T5592

- (2) If the MAIN ADDRESS or the ADDRESS was displayed when the FM was specified in the page name and the MRA3 or MRA4 was used, an incorrect address was sometimes displayed.

Improved ➡ This problem has been fixed.

Relevant revisions: REV 2.00 through REV 6.00B

Relevant systems: T5571P, T5376, T5375, and T5371

3.2.3 Controlling the Handler by using the GPIB Interface (/STHAND)

If the /STHAND command, which selects the ADVANTEST STANDARD setting, and the device test was executed, the following occurred:

- The following DUTs were not tested.

Parallel testing on one station

When testing 64 DUTs in parallel, DUT33 through DUT64

When testing 128 DUTs in parallel, DUT33 through DUT128

When testing 256 DUTs in parallel, DUT33 through DUT256

Parallel testing on two stations

When testing 128 DUTs in parallel, DUT33 through DUT64 and DUT97 through DUT128

When testing 256 DUTs in parallel, DUT33 through DUT128 and DUT161 through DUT256

When testing 512 DUTs in parallel, DUT33 through DUT256 and DUT289 through DUT512

- The HANDLER TYPE setting and the /STHAND command were displayed incorrectly.

Improved ➡ This problem has been fixed.

Relevant revisions: REV 5.03 through REV 6.00B

Relevant systems: T5592, T5586, T5585, T5581, T5581H, T5581P, T5571P, T5376, T5375, T5371, T5336, and T5311

3.2.4 Utility Used to Write Individual Data for Each DUT

If this utility was specified while the PG or TG was activated, no error occurred.

Improved ➡ An error now occurs and an error code (code=20) is stored into the variable, which stores the termination status.

Relevant revisions: REV 5.05 through REV 6.00B

Relevant systems: T5376 and T5375

3.2.5 Voltage Measurement Program (VMEAS)

When VIH, VIL, HVIH, or HVIL was specified to the target unit and the voltage was measured, incorrect results were obtained for pins 1 to 7, 9 to 15, 17 to 23, 25 to 29, 65 and 66.

Improved ➡ Correct results can be obtained.

Relevant revisions: REV 6.00 through REV 6.00B

Relevant system: T5376

3.2.6 Histogram Creation Program (HIDAR)

When a parallel test program, which used two stations, was executed from station 1 (or station 2) and then the histogram for station 2 (or station 1) was displayed by using the .H command, the FILE NAME displayed either nothing or the previous program name.
(The displayed histogram information was correct.)

When this occurred, if the same program was executed from station 2 (or station 1), a runtime error (code=3376) occurred.

Improved ➡ The correct program name is displayed in the histogram.
A parallel test program, which uses two stations, can be executed from station 1 or station 2.

Relevant revisions: REV 1.00 through REV 6.00B

Relevant systems: T5592, T5591, T5591R, T5586, T5585, T5581, T5581H, T5581P, T5571P, T5376, T5375, T5371, T5336, and T5311

3.2.7 Operation Panel

If Lcommand, Ncommand, or Ccommand was displayed on the operation panel and 33 was selected by using the passed-DUT selection button for the Multi DUT Result Pass on station 2, the operation panel terminated abnormally.

Improved ➡ Passed DUTs are displayed correctly.

Relevant revisions: REV 5.01 through REV 6.00B

Relevant systems: T5592, T5591, T5591R, T5586, T5585, T5581, T5581H, T5581P, T5571P, T5376, T5375, T5371, T5336, and T5311

3.3 Controller Command

3.3.1 N Command

Failed test numbers were not displayed under the following conditions:

- A 128 or 256 device parallel test socket program was used.
- S1 or S2 was selected in the N command COMMAND MENU.
- DUT numbers 65 to 96, 129 to 160, and 193 to 224 were selected.

Improved ➡ Failed test numbers are now displayed.

Relevant revisions: REV 5.03 through REV 6.00B

Relevant systems: T5586, T5585, T5376, T5375, and T5371

3.4 Hybrid Controller

3.4.1 Improving of the C++ Program Load

When attempting to load the C++ program, unresolved references to the built-in function generated by the compiler caused an error to occur.

Improved ➡ The C++ program can now be correctly loaded by following the procedure described in Section 1.3.2, “Before Executing C++ Programs” in “ASX/U-51 HYBRID CONTROLLER USER’S MANUAL.”

Relevant revisions: REV 6.00 through REV 6.00B

Relevant systems: T5592, T5586, T5585, T5376, T5375, and T5371

3.4.2 AT_printErrno(), AT_perror(), and AT_strerror()

When an undefined value was specified for errno, an exception occurred and the program sometimes stopped.

Improved ➡ The correct message is now displayed for AT_printErrno(), AT_perror(), and obtained for AT_strerror().

Relevant revisions: REV 6.00 through REV 6.00B

Relevant systems: T5592, T5586, T5585, T5376, T5375, and T5371

3.4.3 atcc

When the -s option was specified, the output filename was created that has a .o extension.

Improved ➡ The output file is now created that has a .s extension.

Relevant revision: REV 6.00B

Relevant systems: T5592, T5586, T5585, T5376, T5375, and T5371

3.4.4 atsymchk

If this command was used to check a C++ program, which had already finished munching, several symbols were treated as defined and they caused an error to occur.

Improved ➡ This problem has been fixed.

Relevant revisions: REV 6.00 through REV 6.00B

Relevant systems: T5592, T5586, T5585, T5376, T5375, and T5371

3.5 ATLworks

3.5.1 ASXDebugger

- (1) If 512 or more characters were used for the structure declaration part of a C program and the variable of the structure was referred to, a GDB segmentation violation error sometimes occurred.

Improved ➡ The structured variable can be now referred to correctly.

Relevant revisions: REV 6.00A through REV 6.00B

Relevant systems: T5592, T5586, T5585, T5376, T5375, and T5371

- (2) If the Step button was clicked in the ATL statements (GOTO CONTINUE or RETURN), which was called from the ATL_exec(), the transfer to the C program by using the Step execution failed.

Improved ➡ This problem has been fixed.

Relevant revisions: REV 6.00A through REV 6.00B

Relevant systems: T5592, T5586, T5585, T5376, T5375, and T5371

- (3) If the program was suspended at the break point, the Reset button on the Control Panel was pressed, and the C program was executed again, the program was not suspended at the break point.

Improved ➡ This problem has been fixed.

Relevant revisions: REV 6.00A through REV 6.00B

Relevant systems: T5592, T5586, T5585, T5376, T5375, and T5371

- (4) If the number of arrays in the structure was omitted as shown below and the variable was referred to, the Member list in the Struct Window was displayed with blank lines. If the Get Data button was clicked on in that situation, a GDB segmentation violation error occurred.

```

struct sample_struct {
    char    data1[16];
    long    data2;
} ;

struct sample_struct data_struct[ ] = {
    { "DATA1", 100 },
    { "DATA2", 200 },
    { "DATA3", 300 }
};

```

↑ The number of arrays is omitted

Improved ➡ The structured variable is now referred to correctly. However, the number of arrays is not clearly defined, so only the first data is displayed.

Relevant revisions: REV 6.00A through REV 6.00B

Relevant systems: T5592, T5586, T5585, T5376, T5375, and T5371

3.5.2 Test Condition Monitor

- (1) Pin Clock monitor and Pin Level monitor

If the socket program, which included the connection definition statement in the following example, and the test plan program, which used the pin list in the following example, were executed, a runtime error (code=3236) sometimes occurred in the pin statement.

(Example)

```

;; Socket program
SOCKET SOC1
:
A1 = PD1 = 33          ;; Pins which were defined by using PDn or
:                    ;; PCn caused an error to occur
END

;; Test Plan Program
PRO PR01 SOC1
:
PINLIST DIN1 = A1      ;; Pin list definition
:
PAUSE                  ;; Pin Clock monitor starts
:                    ;; or refreshes
:
DIN1 = IN1 NRZB BCLK1  ;; Executing a pin statement by using pin list name
:                    ;; causes a runtime error to occur
END

```

Improved ➡ This problem has been fixed.

Relevant revisions: ATLworks REV 1.00 through REV 2.01
REV 5.05 through REV 6.00B

Relevant systems: T5592, T5591, T5591R, T5586, T5585, T5581, T5581H, T5581P,
T5571P, T5376, T5375, T5371

(2) Pin Clock monitor

- (a) When XOR was specified as a waveform mode and BCLK and CCLK were specified as clocks, the MODE item in the pin information frame displayed “-”.

Improved ➡ XOR is now displayed.

Relevant revisions: ATLworks REV 2.00 through REV 2.01
REV 5.05 through REV 6.00B

Relevant system: T5592

- (b) When OPEN was set as the clock timing, “OPEN” was not displayed in the pin information frame or as the timing data in the Setting Pin Clock Window.

Improved ➡ OPEN is displayed.

Relevant revisions: ATLworks REV 1.00 through REV 2.01
REV 5.05 through REV 6.00B

Relevant systems: T5592, T5591, T5591R, T5586, T5585, T5581, T5581H,
T5581P, T5571P, T5376, T5375, T5371

3.6 OS Basic Section

3.6.1 Improving the FATAL TEST SYSTEM ERROR Information

If abnormal temperatures in the test head caused an error to occur, the information on the unit where the abnormal temperature occurred was not displayed correctly.

Improved ➡ This problem has been fixed.

For more information on abnormal temperatures, refer to “T5375 MEMORY TEST SYSTEM TESTER SECTION MAINTENANCE MANUAL” and “T5376 MEMORY TEST SYSTEM TESTER SECTION MAINTENANCE MANUAL.”

Relevant revisions: REV 5.05 through REV 6.00B

Relevant systems: T5376 and T5375